

Market-aware Feature Engineering

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1 Methodology

The 60-feature observation $o_t^{(60)} \in \mathbb{R}^{60}$ used by the team’s PPO baseline is a point-statistic snapshot: per-tick means, last- and last-three-tick views, and episode-historical means of bid, cost, and win-rate signals. It does not encode three classes of information that are recoverable in closed form from the per-tick history cache the simulator already maintains: (i) *time derivatives* of those statistics, (ii) *higher-order moments* of the within-tick competitor-bid distribution, and (iii) *opponent-aggression proxies* that ratio-encode auction outcomes. Since the simulator’s 47 competitors are unobserved, every non-Markov signal the policy might use must be surfaced into o_t explicitly; this extension adds such features at the observation layer while leaving the action space, transition model, reward, and PPO objective untouched.

We define the market-aware observation

$$o_t = [o_t^{(60)}; \phi_t^{(1)}; \phi_t^{(2)}] \in \mathbb{R}^{98}, \quad (1)$$

where $\phi_t^{(1)} \in \mathbb{R}^{20}$ (*Tier 1*) selects 20 named statistics that the simulator’s history cache already maintains per tick but that $o_t^{(60)}$ does not surface (per-slot win counts and exposure rates, raw conversion counts, CPA-exceedence rates, and episode cumulatives), and $\phi_t^{(2)} \in \mathbb{R}^{18}$ (*Tier 2*) is a set of features derived in closed form from the same cache.

The Tier 2 features are computed by applying three standard time-series summaries to per-tick signals already tracked by the simulator. Let $x_{1:t} = (x_1, \dots, x_t)$ denote any such signal (e.g. the within-tick mean of the bid floor lwc_t). With window $W=5$ and $\bar{x} = \frac{1}{W} \sum_k x_{t-W+1+k}$, we use the windowed least-squares slope, windowed sample standard deviation, and episode-history z -score:

$$\text{slope}_W(x_t) = \frac{\sum_{k=0}^{W-1} (k - \bar{k})(x_{t-W+1+k} - \bar{x})}{\sum_{k=0}^{W-1} (k - \bar{k})^2}, \quad \text{std}_W(x_t) = \sqrt{\frac{1}{W} \sum_k (x_{t-W+1+k} - \bar{x})^2}, \quad z_t(x) = \frac{x_t - \bar{x}_{1:t}}{\text{std}_{1:t}(x) + \varepsilon}. \quad (2)$$

Intuitively, slope_W summarises the recent *trend* of x , std_W its recent *volatility*, and z_t how unusual the current tick is relative to the episode’s own history so far. Applied to different underlying signals, these three primitives generate the 18 Tier 2 features, grouped into six categories that target the three blind spots above: (i) *competitor-bid distribution shape* (slopes, stds, and z -scores of lwc , plus right-tail percentile gaps); (ii) *opponent-aggression proxies* (auction-outcome ratios such as lwc_t/p_t and b_t/lwc_t , since opponent bids are unobserved); (iii) *ad-slot dynamics* (top-vs-floor cost and exposure ratios, and their slope); (iv) *regime z -scores* of pv-volume intensity and win rate; (v) *signed momentum* of win rate, ROI p_t/lwc_t , and conversion rate via slope_5 ; and (vi) a

single *budget-pacing residual* $\rho_t = C_t/B - (1 - T_{\text{left}})$ that flags spending ahead of ($\rho_t > 0$) or behind ($\rho_t < 0$) elapsed time. Per-key definitions appear in the Appendix.

The augmented observation is paired with the same clipped-PPO surrogate, the same actor and critic MLP architectures (apart from the input layer width, which grows from 60 to 98), the same 20-period rollout schedule on training periods 7–26, and the same 10M-step training budget. Because the modification is confined to the observation layer, this extension is by construction orthogonal to those that act on the reward (Lagrangian shaping), on rollout sampling (score-weighted reweighting), or on the policy class (GRU temporal encoder, LLM policy), and could in principle be composed with any of them.

2 Results

We evaluate the trained policy on the held-out period 27 under the 48-advertiser *sweep* protocol (one deterministic rollout per advertiser at the advertiser’s true $(B, \text{CPA}_{\text{target}})$). Table ?? reports the held-out sweep score alongside the 16- and 60-feature PPO baselines, all trained for 10M steps with identical PPO hyperparameters and evaluated under the same harness.

Table 1: Held-out period-27 sweep evaluation ($n=48$), mean score $\pm$ s.e.m. All three configurations share PPO hyperparameters, rollout schedule, and the 10M-step training budget; only the observation differs.

Observation	Dim.	Sweep score
PPO baseline (obs_16)	16	20.15 ± 2.25
+ 60-feature state	60	23.24 ± 2.03
+ Market-aware ($o_t^{(98)}$)	98	24.85 ± 2.09

The market-aware observation is the strongest of the three configurations on held-out sweep, improving over the 16-feature baseline by 4.70 score points ($\approx 2.2\sigma$) and over the 60-feature configuration by 1.61 points (suggestive but within one s.e.m.). The ordering is monotone in observation dimension, indicating that each expansion stage—from obs_16 to obs_60, and from obs_60 to the Tier 1 + Tier 2 augmentation—contributes additional non-redundant information that the policy converts into score on the held-out period.

Metrics on constraint-following shed further light. First, the market-aware observation lets the policy sit almost exactly on the target CPA, extracting near-maximal conversion volume without breaching the target, manifested in a realized $\text{CPA}_T/\text{CPA}_{\text{target}}$ ratio of 0.999 for market-aware, compared to 0.884 for obs_16 and 0.991 for obs_60. Cost-over-budget is comparable across the three configurations (0.94–0.95), so the score gap is driven by better CPA-aware bid *shaping* rather than by an overall change in how much budget is spent. The takeaway is that the additional features pay off through finer constraint-aware allocation, not through increased throughput, and the gain accrues monotonically with observation dimension on held-out sweep.

Two qualifications temper this read. The decisive obs_60 $\rightarrow$ market-aware step is +1.61 points, within one s.e.m. of either baseline at $n=48$ and a single seed; the result is suggestive but does not separate from the null. The augmentation also bundles three feature categories (time derivatives, distribution shape, opponent proxies) and two implementation tiers (free selection vs. derived features) into a single observation, so the gain cannot yet be localised to a specific source—most plausibly the opponent-aggression proxies, given that the largest movement is in CPA compliance.

The training trajectory is consistent with the held-out outcome. The per-rollout score logged during training rises from ≈ 18.1 averaged over the first 1M steps to ≈ 25.4 averaged over the final 1M steps, with cost-over-budget ≈ 0.93 and target-cpa-over-cpa ≈ 1.05 at convergence. The training-distribution score plateaus around 3M steps and remains flat through 10M, so the budget is sufficient and the train-to-test gap (≈ 0.5 points) is small.

Two follow-ups are natural. First, a permutation-importance analysis on the trained policy—shuffling each added key (or each Tier 2 category as a block) at evaluation time and measuring the resulting drop in held-out sweep score—would reveal which categories of feature drive the majority of the improvement and which are redundant with signal already present in $o_t^{(60)}$. The resulting ranking would let us distill $o_t^{(98)}$ to a leaner subset that retains the score gain while reducing input dimensionality, easing optimisation and clarifying the mechanism behind the held-out improvement. Second, because this extension is confined to the observation layer, it is by construction composable with the team’s other interventions; stacking it with reward-shaping (λ -CPA Lagrangian) is the most promising pairing, since the two act on the same constraint along orthogonal axes (state representation versus reward signal).

3 Appendix

A.1 Windowed primitives

For any tracked scalar signal $x_{1:t}$ and window length W , with $\bar{k} = (W-1)/2$ and $\bar{x} = \frac{1}{W} \sum_{k=0}^{W-1} x_{t-W+1+k}$,

$$\text{slope}_W(x_t) = \frac{\sum_{k=0}^{W-1} (k - \bar{k})(x_{t-W+1+k} - \bar{x})}{\sum_{k=0}^{W-1} (k - \bar{k})^2}, \quad \text{std}_W(x_t) = \sqrt{\frac{1}{W} \sum_{k=0}^{W-1} (x_{t-W+1+k} - \bar{x})^2}. \quad (3)$$

The slope is the closed-form OLS estimate against an integer time index; the episode z -score $z_t(x) = (x_t - \bar{x}_{1:t})/\text{std}_{1:t}(x)$ uses cumulative mean and std over the episode prefix, with $\text{std}_{1:t}$ floored at a small ε to guard against constant prefixes.

A.2 Tier 1 surfaced statistics (20 keys)

Per-slot win counts $\{\text{bid_success_count_slot_}j\}_{j=1,2,3}$ in last/last-three/historical views; per-slot exposure rates $\{\text{exposure_mean_slot_}j\}_{j=1,2,3}$ in the same three views; last-three and historical raw conversion counts; last/last-three/historical CPA-exceedence rates; and the episode cumulatives Y_t, C_t, CPA_t . These quantities are already maintained by the simulator’s per-tick history update, so Tier 1 is implemented as a selection-only change to the JSON observation key list with no code change to the environment.

A.3 Tier 2 derived features (18 keys)

Below lwc_t denotes the within-tick mean of the least winning competitor cost (the auction floor faced by the agent), b_t the agent’s mean submitted bid at tick t , p_t the mean predicted conversion probability over the tick’s impression batch, wr_t the win rate, and C_t, B, T_{left} the cumulative spend, total budget, and remaining time fraction.

- **Competitor distribution shape (7).** $\text{slope}_5(\text{lwc})$ and $\text{std}_5(\text{lwc})$ capture trend and volatility of the bid floor; the coefficient of variation $\text{std}_5/\text{mean}_5$ separates absolute from relative volatility; $z_t(\text{lwc})$ locates the current tick within the episode’s own scale; the right-tail width

$\text{lwc}_t^{(01\%)} - \text{lwc}_t^{(10\%)}$ and ratio $\text{lwc}_t^{(\max)}/\text{lwc}_t^{(\text{mean})}$ expose “whale” bidders whose presence the baseline mean smooths out; slope_5 of the percentile gap tracks whether the tail is widening.

- **Opponent-aggression proxies (2).** Because competitor bids are unobservable, we infer opponent stance from auction outcomes: $\alpha_t^{\text{opp}} = \text{lwc}_t/p_t$ (price the market clears per unit of predicted conversion) and the agent’s own competitiveness $\beta_t = b_t/\text{lwc}_t$.
- **Slot dynamics (3).** The cost ratio $c_t^{(1)}/c_t^{(3)}$ between the top and floor ad slot, the analogous exposure ratio, and the slope_5 of slot concentration. Together they signal whether premium ad real estate is being contested or vacated.
- **Regime z -scores (2).** z_t of pv-volume intensity and of win rate, locating the current tick within the episode’s empirical distribution rather than against an absolute scale.
- **Momentum / acceleration (3).** slope_5 of wr_t , of the ROI proxy p_t/lwc_t , and of the per-tick conversion rate.
- **Budget-pacing residual (1).** $\rho_t = C_t/B - (1 - T_{\text{left}})$, with $\rho_t > 0$ indicating cumulative spend ahead of elapsed time and $\rho_t < 0$ indicating underpacing.